A Held-Out Audit of Nonlinear Filters for Light-Level Geolocation of a Juvenile White Shark

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***Abstract*—Pop-up archival transmitting (PAT) tags infer animal position from the timing of sunrise and sunset. The measurement is well conditioned in longitude and nearly singular in latitude near an equinox, which makes it a useful stress test for nonlinear filtering. I build an extended Kalman filter, an unscented Kalman filter, a bootstrap particle filter, two smoothers and a grid HMM on one 137-day juvenile white shark (*Carcharodon carcharias*) record, then freeze that configuration and apply it unchanged to six further double-tagged animals from the same public archive. Every animal is scored against Argos fixes from a second tag on the same shark. The contribution is an audit rather than an error number: the final 30 days of the development animal are held out before any model code is written, the other six animals are never tuned on, and every stochastic run is repeated over three to ten seeds. Five results follow. *(1) The cohort is where the real defects were.* Six new animals exposed two faults one animal could not reveal: a seafloor constraint keyed to the tuned animal's timestamps, which silently disabled itself on every other shark, and a filter clock anchored to the tuned animal's release date, which froze one animal on its release prior for its entire record and gave it the lowest error in the study. With both repaired, halving the process noise helps all seven animals, by 8.5 to 32.4% (Table I, Fig. 4). *(2) The largest transferable gain was data handling, not estimation.* The temperature reference field had been downloaded for the tuned animal's deployment window only, so 149 of the cohort's 1052 dawn readings had anything to compare against and three animals had none at all; the likelihood also read a missing reference as an impossible position rather than as no information. Repairing both improves all seven animals, by 2.6 to 52.3%. The two repairs together cut the cohort mean of per-animal medians from 282.2 to 171.0 km (Section XIV). *(3) One constant pools across animals and one does not.* A temperature offset set leave-one-animal-out beats the single-animal offset on all seven (sign test $p = 0.008$) and reaches 162.2 km, against 162.8 km for an oracle fitted on each animal's own fixes. The same treatment of the process-noise scale collapses, from 141.8 to 727.1 km on the animal that set the shipped value, and a particle-budget ladder attributes most of that cliff to sampler convergence rather than biology (Section XV). *(4) Tuning-window numbers did not generalise.* My best single-animal configuration reaches 117 km on the window I tuned on and 544 km on the held-out window (Table II). Seed-to-seed spread is 29–90 km, far larger than the 1–5 km steps an earlier ablation ladder reported, and across 75 variants tuning rank predicts held-out rank only weakly ($\rho = +0.30$), with the best-on-tuning variant ranking 54th of 75 (Section VI-E). *(5) The dominant error has a cause no filter in this family can remove.* A twilight pair is two numbers and position is two unknowns, so a threshold bias is confounded with position by construction: one minute of common-mode bias is observationally identical to 25 km of longitude. A bias fix chosen without touching held-out data improves calibration and not accuracy, and widening the family makes it worse (Section IX); a particle smoother, which adds no parameter, cuts tuning-window error from 119 to 69 km while held-out error is statistically unchanged (Section X), confirming that what remains is bias rather than variance. GPE3, the manufacturer's proprietary HMM smoother, is the only estimator here that holds up on both windows. Scored by the same code on the same fixes it is ahead on all seven animals, 52.1 against 162.2 km, so I use it throughout as an external benchmark rather than a target I claim to beat.**

***Index Terms*—state estimation, particle filter, unscented Kalman filter, animal telemetry, model validation, overfitting, reproducibility**

I. Introduction

A PAT tag carries no receiver. It records light, pressure and temperature, and position must be inferred afterwards from the astronomy of the light curve. Local apparent noon gives longitude directly and robustly. Latitude comes only from day length, and $\partial(\text{day length})/\partial(\text{latitude}) \to 0$ at an equinox, so for weeks around it the inverse problem is locally rank deficient in latitude. This deployment runs straight through the March 2008 equinox.

That makes it a good test case for nonlinear filtering, and the second tag on the same animal makes it a rare one: the Argos fixes give an independent answer that the light data never sees. Most light-level geolocation papers validate against simulation or against a single reference track. Here there is a real, external, irregularly sampled ground truth.

This paper is deliberately an audit rather than a method paper. An earlier version of this pipeline reported a best median error near 119 km and a ladder of incremental improvements. Everything in Section V was added afterwards, and most of what the ladder claimed did not survive it. The useful output is therefore a set of statements about what the data can and cannot resolve, and a short list of process errors that produced the original over-claim.

II. Data

A. Tag record

Deployment 07_05 is a juvenile white shark tagged off central California on 5 February 2008. The archival tag was physically recovered, so the full-resolution record is available rather than the summarised transmission: light level, pressure and temperature at 10-second to 1-minute cadence over 137 days. The animal left the shelf and moved south and offshore, ending near the southern Gulf of California.

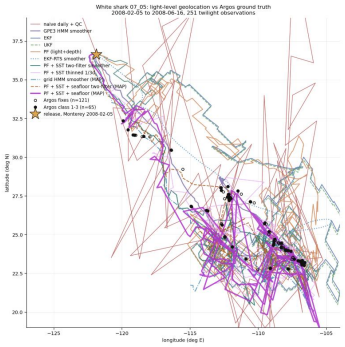


Fig. 1. Argos ground truth (black) and estimated tracks. The animal leaves the California shelf, crosses the equinox offshore, and ends in the southern Gulf of California.

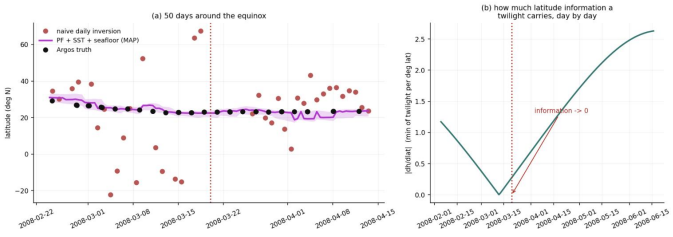


Fig. 2. Sensitivity of day length to latitude through the deployment. The derivative passes through zero at the March equinox, where latitude is unobservable from day length at any noise level.

Twilight events are extracted in two ways. The manufacturer's LightLoc processing yields 251 twilight times, 241 after quality control. A re-extraction directly from the raw light curves yields 228 events with an explicit depth of detection attached to each, which Section VII-C uses.

B. Ground truth

A SPOT tag on the same animal returned 122 Argos fixes between 15 February and 15 June 2008. Location classes are 3 ($n = 3$), 2 (20), 1 (42), 0 (24), A (16) and B (16). I assign them nominal one-sigma radii of 0.25, 0.5, 1.5, 10, 3 and 9 km respectively; classes 0, A and B are unquoted by Argos, so these are the empirical marine-vertebrate values rather than anything the system guarantees.

The Argos track is the only ground truth used anywhere in this paper. It is never an input to any estimator. Fig. 1 shows it together with the estimated tracks.

III. Observation and Motion Models

A. Twilight timing

For a twilight event at time t of kind $k \in \{\text{dawn}, \text{dusk}\}$, the predicted event time is obtained by solving for the instant at which the solar elevation equals a threshold $h_0^{(k)}$. The threshold is not the geometric horizon: it absorbs sensor

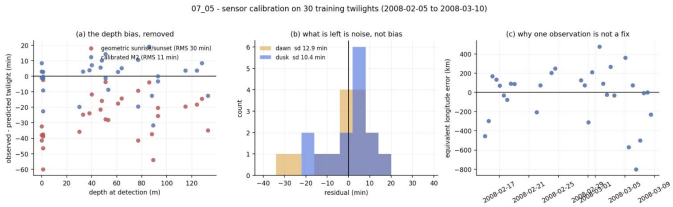


Fig. 3. Calibration-window twilight residuals after fitting (1) on the LightLoc times. The fit is unbiased on the window it was fitted to, which is the point: nothing here predicts the behaviour in Table V.

response, cloud, and the fact that the tag may be at depth when the threshold is crossed. I model it as

$$h_0^{(k)}(z) = h_0^{(k)} + \beta \log(1 + z), \tag{1}$$

with z the depth in metres at which the twilight was recorded. The residual is taken in minutes and given a Student-t likelihood with $\nu = 4$ degrees of freedom, which matters more than any other modelling choice here (Section VI-G).

Two calibrations of (1) appear in this paper. The first is frozen on the LightLoc times in the calibration window and gives $h_0^{\text{dawn}} = -9.91°$, $h_0^{\text{dusk}} = -0.29°$ and $\beta = +0.89$ per log-metre, with per-branch residual standard deviations of 12.88 and 10.43 minutes. The second refits the same form on the raw-light times and gives $h_0^{\text{dawn}} = -5.56°$, $h_0^{\text{dusk}} = -5.88°$ and $\beta = +0.0085$, with a residual RMS of 4.80 minutes. The second looks much better and behaves much worse; Section VII-C is about that. Fig. 3 shows the first fit's residuals on the window it was fitted to.

B. Sea-surface-temperature aiding

Daily 0.25° optimally interpolated SST is compared with the tag's shallowest daily temperature, and the resulting log-likelihood is added to the twilight term. The point of SST is not that it is accurate. It is that its information is orthogonal to day length, so it carries latitude through the equinox where the light data cannot. Fig. 2 shows the sensitivity of day length to latitude passing through zero there. In March alone it moves the particle filter from 522 to 109 km median error.

C. Seafloor-depth aiding

Where the tag's maximum daily depth exceeds the ETOPO seafloor depth at a candidate position, that position is impossible. This enters as $\log \Phi((d_{\max}(p) - z)/\sigma_B)$ with $\sigma_B = 150$ m, plus a finite land penalty. It is a hard, highly informative constraint, and it makes the posterior strongly multimodal, which is why the choice of point estimate stops being innocent.

D. Motion model

The state is $x = [\varphi, \lambda, v_E, v_N]$ with a damped random walk in velocity,

$$\dot{v} = -v/\tau + w, \qquad \tau = 3 \text{ days}, \tag{2}$$

and σ_v set so the stationary speed distribution has a sensible mean for a juvenile white shark. Positions are confined to a generous box (15°–50° N, 140°–105° W).

IV. Estimators

I implement, all on the same observation and motion models: a naive event-by-event inversion as a floor; an EKF and a UKF with 3σ innovation gating; Rauch–Tung–Striebel and unscented RTS smoothers for each; a bootstrap particle filter with systematic resampling and roughening; a two-filter particle smoother; a cloud-level fusion of two particle filters differing only in their observation model; and a grid HMM over a $0.25°$ lattice. The proprietary GPE3 hidden-Markov smoother supplied by the tag manufacturer is included as an external reference. It is proprietary, so I can score against it but not inspect it.

V. Evaluation Protocol

This section is the part of the paper I would read first, because every number after it depends on it and none of it existed in the earlier version.

A. Three windows

The record is split once, by date, and never re-split:

- **Calibration**, up to 10 March 2008. The only window in which any sensor parameter is fitted.
- **Tuning**, 11 March to 15 May 2008. Every modelling decision, hyperparameter and readout choice in this paper was made by looking at this window. It contains 64 truth-matched twilight events.
- **Lockbox**, the final 30 days from 16 May 2008. Scored once per configuration, after the configuration was frozen. It contains 20 events.

The lockbox is small, and the consequence is worth stating before any number rests on it. With 24 matched Argos fixes the seed standard deviation alone runs 80–90 km (Tables III and XVII), so nothing smaller than roughly 170 km—over half the lockbox baseline—clears twice the spread, while the tuning window resolves about 13 km on the same basis. The lockbox can detect large effects and cannot adjudicate moderate ones. That is a property of the held-out set I designed, not of any result scored on it, and Section XV is where it bites.

Nothing is selected on the lockbox anywhere, including the one hyperparameter introduced in Section VIII. Where a lockbox column appears next to a selection rule, it is printed for the reader and was not available to the rule.

B. Metrics

Error is the great-circle distance between an estimate interpolated to an Argos fix time and that fix, reported as a median over matched events; the median rather than the mean because the naive inversion produces occasional thousand-kilometre outliers that would otherwise dominate. Alongside it I report 1σ coverage: the fraction of matched events at which the truth falls inside the estimator's own reported standard deviation. Nominal coverage for a Gaussian is 68%. An estimator that is wrong is usable; an estimator that is wrong and confident is not.

C. Monte Carlo repetition

Every particle-based result is the mean over ten seeds, reported with its standard deviation across seeds. This is the single cheapest change in the paper and the one that invalidated the most prior conclusions.

VI. Results

A. The same pipeline on seven animals

Table I and Fig. 4 are the headline result, so they come before anything measured on a single animal: seven animals, one frozen pipeline, no per-animal tuning. The table is the cumulative effect of the three interventions that survived the audit, each applied to the frozen configuration and scored against every animal's own Argos fixes: halving the process noise once two cohort-only defects were repaired (Section XIII), extending the temperature reference field over the whole cohort and correcting its missing-data rule (Section XIV), and replacing the single-animal temperature offset with one pooled leave-one-animal-out (Section XV). Only 07_05 was ever tuned on; for the other six every day is held out.

Every animal improves, by 26 to 67%, and the cohort mean of the per-animal medians falls from 282.2 to 162.2 km. Two things about that are worth saying plainly. None of it comes from a better estimator: the three columns are a motion-model scale that had been fitted on the wrong month, a reference file that covered one deployment window out of seven, and a constant averaged over six animals instead of one. And it is still not competitive with the external benchmark. The archive carries a GPE3 track for every animal, so I scored it with the same code on the same fixes: it is ahead on 7 of 7, by factors of 1.8 to 7.6, and its cohort mean of per-animal medians is 52.1 against my 162.2 km. Fig. 4 draws both against the fixes.

B. Where the methods land on the development animal

Everything from here to Section XIII is diagnostic work on 07_05, the one animal I was allowed to tune on, and Table II is where its estimators land. It gives median error on the full evaluation set and, separately, on the tuning window and the lockbox. All rows use one scorer and one set of matched events, so the columns are comparable.

Two things are worth reading off it immediately. The GPE3 smoother is better than everything I built, at 72 km, and that gap is not closed anywhere in this paper. And the ordering of my own rows changes completely between the tuning column and the lockbox column: the best tuning-window configuration (84.9 km) is nearly the worst lockbox configuration (543.9 km), while a plain EKF smoother is the best lockbox configuration at 319 km. Fig. 5 separates the two coordinates: longitude is recovered by everything, and the rows differ almost entirely in latitude.

TABLE I

THE COHORT IN ONE TABLE. MEDIAN GREAT-CIRCLE ERROR IN km AGAINST EACH ANIMAL'S OWN ARGOS FIXES, ACCUMULATED LEFT TO RIGHT. PUBLISHED IS THE FROZEN 07_05 CONFIGURATION WITH THE TWO COHORT DEFECTS REPAIRED; $\sigma_v/2$ HALVES THE PROCESS NOISE; +SST ADDS THE EXTENDED REFERENCE FIELD AND THE CORRECTED MISSING-DATA RULE; +POOLED REPLACES THE SINGLE-ANIMAL TEMPERATURE OFFSET WITH A LEAVE-ONE-ANIMAL-OUT ONE. THREE SEEDS; SEED SPREADS ARE GIVEN PER COLUMN IN TABLES XV, XVI AND XVIII. GPE3 IS THE MANUFACTURER'S SMOOTHER SCORED BY THE SAME CODE ON THE SAME ARGOS FIXES, SO THE LAST COLUMN IS A LIKE-FOR-LIKE RATIO OF +POOLED TO GPE3. † MARKS THE ONLY ANIMAL EVER TUNED ON.

Animal	Fix	Published	$\sigma_v/2$	+SST	+pooled	Total	GPE3	Ratio
07_05†	115	364.4	333.4	159.1	141.8	−61.1%	79.6	1.78
07_03	140	526.3	477.0	401.5	388.0	−26.3%	51.0	7.62
08_01	47	166.3	130.4	80.6	78.1	−53.0%	31.2	2.51
08_02	102	162.7	135.9	107.6	104.9	−35.5%	38.8	2.70
08_04	148	234.4	180.9	176.2	159.8	−31.8%	62.5	2.56
08_09	139	308.8	233.3	199.9	192.5	−37.7%	63.0	3.06
09_15	18	212.8	143.8	72.2	70.4	−66.9%	38.5	1.83
Mean		282.2	233.5	171.0	162.2	−42.5%	52.1	3.12

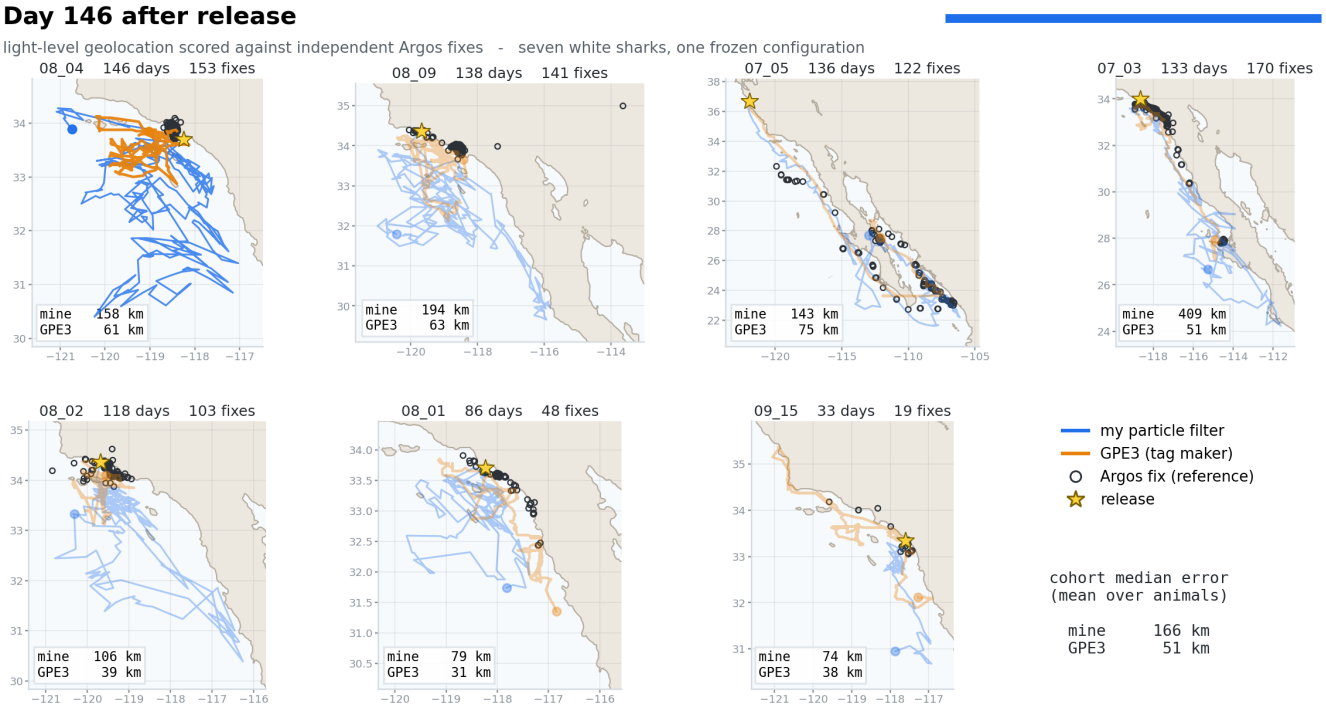


Fig. 4. The cohort at the end of deployment. Grey rings are each animal's own Argos fixes, orange is the GPE3 reference track, blue is my filter in the configuration of Table I's +pooled column at a single seed. Panels are ordered by deployment length; the running medians printed in each panel are over the fixes drawn. Only 07_05 was ever tuned on. The repository holds an animated version of this figure, one frame per 1.2 days.

C. The headline was a tuning-window number

The earlier version of this work quoted its best configuration at roughly 119 km and stopped there. That number survives re-running: the two-filter cloud fusion scores 117.4 km on the full evaluation window. What does not survive is the interpretation. On the final 30 days the same configuration scores 543.9 km, against 118.4 km for GPE3 on exactly those days. The full-window figure is dominated by the tuning window, which carries 64 of the 84 matched events, so quoting it as an accuracy is close to quoting a training error.

D. Seed noise sets the resolution floor

Table III repeats one configuration ten times at 120,000 particles. The standard deviation across seeds is 29 km on the full window and 90 km on the lockbox. The earlier ablation ladder reported improvements of 1–5 km per step and treated them as real. They are an order of magnitude below this floor. Measuring the noise before building the ladder would have removed most of the ladder.

E. Tuning-window skill barely transfers

Every pipeline variant I built was scored on both windows: 75 configurations, spanning filters, smoothers, aiding combinations, readouts and particle counts. Fig. 6 plots lockbox

TABLE II

MEDIAN GREAT-CIRCLE ERROR (km), ONE SCORER AND ONE MATCHED EVENT SET THROUGHOUT. "TUNE" AND "LOCK" ARE THE TUNING WINDOW AND THE HELD-OUT FINAL 30 DAYS. GPE3 IS THE MANUFACTURER'S SMOOTHER, INCLUDED AS AN EXTERNAL REFERENCE.

Method	All	Tune	Lock
Naive daily inversion	513.8	570.5	473.7
Naive daily inversion, QC	478.8	487.0	473.7
EKF	347.2	285.2	413.0
UKF	356.6	291.2	417.1
PF, light + depth	316.9	240.2	391.0
EKF + RTS smoother	285.7	169.4	359.0
UKF + unscented RTS	283.7	173.0	362.9
EKF + RTS, QC events	279.9	169.5	**319.4**
PF + SST	198.4	114.9	498.4
PF + SST, thinned 1/3 d	186.1	129.2	473.1
Grid HMM filter	163.9	130.1	569.2
PF + SST, two-filter smoother	155.2	112.2	422.2
PF + SST + seafloor	152.8	123.2	452.4
Two-filter cloud fusion (my best)	117.4	**84.9**	543.9
GPE3 HMM smoother (external)	72.0	71.1	118.4

TABLE III

TEN SEEDS OF ONE CONFIGURATION AT 120,000 PARTICLES, WITH AND WITHOUT THE THRESHOLD-BIAS STATE OF SECTION VIII. ERRORS ARE MEDIANS IN km; COV IS 1σ COVERAGE AGAINST THE FILTER'S OWN REPORTED STANDARD DEVIATION; $\bar{s}$ IS THE MEDIAN REPORTED STANDARD DEVIATION IN THE LOCKBOX.

	All	Tune	Lock	cov_T	cov_L	$\bar{s}_\text{L}$
As published	408.1	363.4	532.4	22%	9%	201
± seed sd	29.2	31.0	89.5			
+ bias state	420.4	358.1	615.4	38%	8%	205
± seed sd	17.7	13.1	35.0			

error against tuning error for all of them. The Spearman rank correlation is $\rho = +0.30$ ($p = 8.8 \times 10^{-3}$).

A weak positive correlation is worse news than it sounds. It means tuning-window rank carries some information about held-out rank, but far too little to select on. Concretely: the configuration that looks best while tuning (84.9 km) comes 54th of 75 on the lockbox, and the configuration that is actually best on the lockbox (319.4 km, an EKF smoother on QC'd events) comes 40th of 75 on tuning. Picking the tuning-window winner is close to picking at random, and because the search was large, the winner is mostly the variant that best fitted the noise in 64 events.

F. The filters do not know they are wrong

Coverage is where the failure is least ambiguous. In Table III the published configuration attains 22% 1σ coverage on the tuning window and 9% on the lockbox, against a nominal 68%. In the lockbox it reports a median standard deviation of 201 km while being wrong by 532 km — roughly a 2.6σ systematic offset, sustained over 30 days, reported as if it were a 1σ situation.

Under-coverage this severe is not a tuning problem. A filter whose process noise is merely too small produces errors that are too large relative to its variance in a random direction.

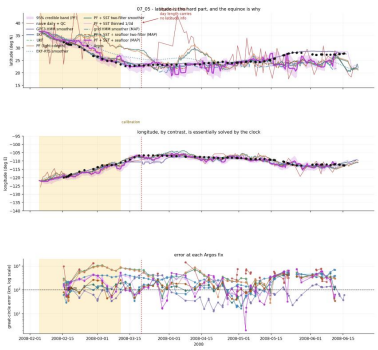


Fig. 5. Latitude and longitude against time for the main estimators. Longitude is recovered well by everything; latitude is where the estimators separate, and where the systematic offset of Section VII-C accumulates.

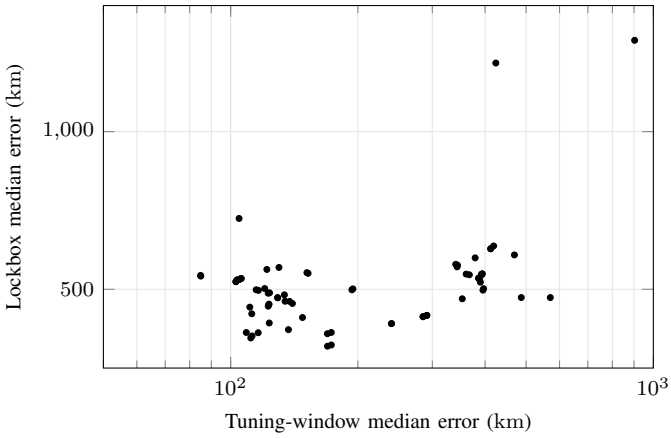


Fig. 6. Every pipeline variant I built ($n = 75$), scored on the window I tuned on (horizontal) and on the held-out window (vertical). If tuning-window skill transferred, the cloud would run along the diagonal. Spearman $\rho = +0.30$.

Here the error has a consistent sign in latitude, which points at the measurement model rather than at the filter.

G. A matched comparison of EKF, UKF and PF

The particle filter beats the Kalman filters in Table II, but it also differs from them in two ways at once: it uses a Student-t likelihood and it enforces the feasibility box. Table IV separates these by running all four combinations through the same particle filter code path.

The heavy tails are worth 101 km; the box is worth 9 km. A Gaussian particle filter is 84 km *worse* than the EKF, despite representing the posterior far more faithfully than a covariance can. So the honest statement is not that the particle filter beats the Kalman filters. It is that a Student-t likelihood beats a Gaussian one, and either filter can have that. The non-Gaussian posterior was not the point.

VII. DIAGNOSIS

A. The residual at the true position

Because the Argos fixes exist, I can do something better than inspect filter output: I can evaluate the measurement model at the *true* position. For each twilight event with an Argos fix

TABLE IV

MATCHED ABLATION OF THE PARTICLE FILTER'S ADVANTAGE. THREE SEEDS AT 20,000 PARTICLES, ALL ROWS THROUGH THE SAME CODE PATH ON THE SAME EVENTS.

Configuration	All	Tune	Lock
PF, Student-t $\nu = 4$, box	321.2	238.3	395.9
PF, Student-t $\nu = 4$, no box	330.2	273.7	396.4
PF, Gaussian, box	413.7	384.1	460.5
PF, Gaussian, no box	430.9	389.8	478.7
EKF	347.2	285.2	413.0
UKF	356.6	291.2	417.1

TABLE V

BRANCH-CORRECTED TWILIGHT RESIDUAL AT THE TRUE POSITION, BY HALF-MONTH. y IS IN MINUTES, DEPTH IN m, AND $|z|$ IS THE RESIDUAL IN UNITS OF THE MODEL'S OWN σ.

Half-month	n	y	depth	$\lvert z\rvert$
2008-02a	1	-10.3	88	2.9
2008-02b	9	-10.1	48	5.0
2008-03a	7	$+34.9$	51	6.1
2008-03b	7	$+33.9$	13	5.9
2008-04a	6	$+34.8$	50	6.1
2008-04b	9	$+33.9$	41	5.9
2008-05a	11	$+55.0$	56	9.6
2008-05b	6	$+27.7$	94	5.7
2008-06a	7	$+13.5$	92	3.8
Tuning	51	$+34.2$	49	—
Lockbox	12	$+27.5$	98	—

within 8 hours (63 of 228 events), I compute the predicted twilight time at the interpolated true position and subtract the observed one. This isolates measurement-model error from filter error completely.

Table V gives the result by half-month. Two features matter. The branch-corrected residual y sits near $+34$ minutes from the first half of March onwards, which is five to six times the model's own σ; and the two twilight branches move in *opposite* directions, with lockbox medians of $+12$ minutes at dawn and -71 minutes at dusk.

The antisymmetry is the diagnostic. A clock error, or a drift in the tag's time base, would shift both branches the same way. An antisymmetric shift compresses the apparent day length, and day length is exactly what carries latitude. At this time of year about 60 minutes of day-length error is worth roughly $650\,\mathrm{km}$ of latitude, which matches the size and the sign of the lockbox latitude bias of -413 to $-543\,\mathrm{km}$ measured in Table VII.

So the model was not slightly wrong at the end of the deployment. It was wrong by five to six of its own sigmas for the whole record, and the lockbox merely made that visible.

B. Depth of detection, and an unidentifiable parameter

The depth at which a twilight is recorded is the obvious candidate mechanism: light attenuates with depth, so a tag deeper in the water column crosses any fixed irradiance threshold later at dawn and earlier at dusk — antisymmetric, as observed.

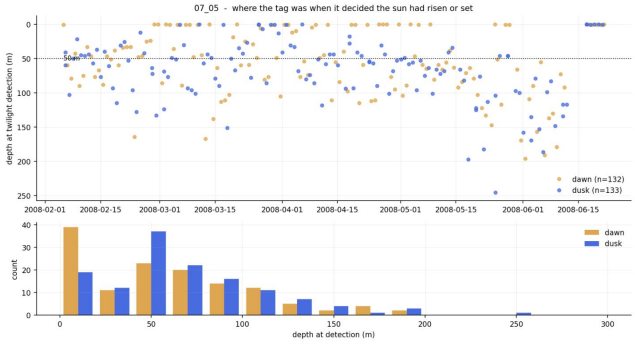


Fig. 7. Depth at which each twilight was recorded, through the deployment. The calibration window reaches $94\,\mathrm{m}$ at the 95th percentile; the lockbox reaches $182\,\mathrm{m}$. The depth term in (1) is fitted entirely inside the shallower range.

The depth distribution differs sharply between windows (Fig. 7). In the calibration window the median twilight depth is $51\,\mathrm{m}$ and the 95th percentile is $94\,\mathrm{m}$; in the lockbox the median is $98\,\mathrm{m}$ and the 95th percentile is $182\,\mathrm{m}$. The parameter β in (1) was therefore fitted over a range of depths that the lockbox almost entirely exceeds.

Quantitatively the mechanism is consistent but weakly evidenced. Regressing y on $\log(1+z)$ over the non-lockbox events gives $y \approx 8.8 + 4.9\log(1+z)$, and extrapolating that to the lockbox median depth of $98\,\mathrm{m}$ predicts $+31.5$ minutes against $+27.5$ observed. The magnitude is right. But the rank correlation itself is weak and not significant ($\rho = +0.27$, $p = 0.05$, $n = 51$), so I would call the mechanism consistent with the data rather than established by it.

The sharper statement is about identifiability. Fitted on the raw-light twilight times, β comes out at $+0.0085$ per log-metre, essentially zero. Fitted on the LightLoc times, the same physical quantity comes out at $+0.89$. Two orders of magnitude apart. No goodness-of-fit diagnostic in the calibration window flags this, because the calibration window does not span enough depth to constrain β at all. A support check does flag it, and costs nothing.

C. A recalibration that made things worse

This is the single most useful negative result here, and the easiest to state.

Refitting (1) on the raw-light twilight times cuts the calibration-window residual RMS to 4.80 minutes, against per-branch standard deviations of 12.88 and 10.43 minutes for the original LightLoc calibration. By every calibration-window diagnostic it is a large improvement.

Held out, it is a large regression. On identical events, the identical filter and the identical seed, the refitted constants give $364.4\,\mathrm{km}$ median error on the evaluation window where the original constants give $124.3\,\mathrm{km}$. A recalibration that more than halves the residual it was fitted to makes out-of-sample accuracy roughly three times worse.

The mechanism is the one above: the refit buys its low residual by driving β to zero, which is defensible inside

TABLE VI
SELECTING THE THRESHOLD-BIAS RANDOM-WALK SCALE BY TUNING-WINDOW MARGINAL LIKELIHOOD. THREE SEEDS AT 20,000 PARTICLES. THE LOCKBOX COLUMNS WERE PRINTED AFTER THE SELECTION WAS MADE AND PLAYED NO PART IN IT.

σ_b	$\log Z_{\text{tune}}$	Tune	cov_{T}	Lock	cov_{L}
0	-1176.2	315.6	27%	538.8	15%
1	-1087.7	260.6	49%	650.2	3%
2	-1069.5	330.9	40%	462.2	28%
3	$\mathbf{-1060.0}$	389.8	33%	567.8	10%
5	-1061.8	324.4	34%	423.6	18%
8	-1070.3	414.8	27%	589.0	5%
12	-1073.3	427.1	18%	720.3	5%

a window spanning 0–94 m and indefensible in a window reaching 182 m. The general lesson is that a sensor-error model fitted on a window that does not span the operating range of its own regressors will look well calibrated on that window and fail silently outside it.

A practical note belongs here too. In the notebook these constants live at module scope and are installed by a context manager, because the first version of this audit ran the raw-light filters outside that scope and silently used the wrong constants — producing numbers that looked three times better than they were. The audit now begins with an equivalence test asserting that the bias-state filter with $\sigma_b = 0$ reproduces the published filter to within seed noise (-8.3 and -6.4 km on the two readouts). Global mutable calibration state is a reproducibility hazard, and a cheap assertion catches it.

VIII. A Fix Chosen Without the Lockbox

The diagnosis says the threshold is biased and antisymmetric. Rather than try to predict that bias offline — which Section VII-C shows is exactly where the previous attempt failed — the filter can estimate it online. I add one scalar to the state, a threshold bias b in minutes entering the observation as $+b$ at dawn and $-b$ at dusk, evolving as a random walk with scale σ_b per $\sqrt{\text{day}}$. The state becomes $[\varphi, \lambda, v_E, v_N, b]$. This adds one hyperparameter and no new data.

A. Selecting σ_b honestly

σ_b is chosen by maximising the filter's own marginal likelihood accumulated over the tuning window only. Table VI shows the sweep. The evidence criterion selects $\sigma_b = 3$. For comparison, a rule that matched tuning-window coverage to nominal would have selected $\sigma_b = 1$, and the value that is actually best on the lockbox is $\sigma_b = 5$ — which the evidence rule did not pick, as it should not, having never seen the lockbox.

Of the six non-zero values, all six beat $\sigma_b = 0$ on tuning-window evidence, two beat it on lockbox median error, and two beat it on lockbox coverage. The evidence criterion reliably prefers a bias state; the held-out data does not reliably reward one.

TABLE VII
CANDIDATE RESPONSES TO THE DIAGNOSIS, THREE SEEDS AT 20,000 PARTICLES. $\bar{s}_{\text{L}}$ IS THE MEDIAN REPORTED STANDARD DEVIATION AND $\Delta\varphi_{\text{L}}$ THE MEDIAN LATITUDE BIAS, BOTH IN THE LOCKBOX.

	Tune	Lock	cov_{L}	$\bar{s}_{\text{L}}$	$\Delta\varphi_{\text{L}}$
As published	315.6	538.8	15%	219	-413
Twilight $\sigma \times 2$	220.5	528.4	15%	190	-473
Twilight $\sigma \times 3$	164.6	555.9	5%	206	-548
Depth QC, $z \le 94$ m	478.1	978.6	0%	178	-863
Bias state, $\sigma_b = 2$	330.9	462.2	28%	196	-424
Bias state, $\sigma_b = 3$	389.8	567.8	10%	176	-543
Depth QC + bias state	460.1	921.5	0%	214	-838

B. What the fix buys, and what it does not

Table III gives the honest answer at full particle count over ten seeds. Adding the bias state moves tuning-window 1σ coverage from 22% to 38%, a real and reproducible gain. It leaves the full-window error unchanged ($+12.3$ km, 1.1 standard errors), leaves tuning-window error unchanged (-5.3 km, 0.5 standard errors), and makes lockbox error *worse* ($+83.0$ km, 2.7 standard errors). Lockbox coverage stays flat at 8–9%.

So the fix is the right response to the diagnosis, it is selected without touching held-out data, and it does not generalise. I am reporting it because it failed. Had I selected σ_b on the lockbox I could have reported 423.6 km and a 109 km improvement, and it would have been meaningless.

Table VII puts the fix beside the two alternatives one would naturally reach for. Inflating the twilight σ improves the tuning window substantially ($316 \to 165$ km at $3\times$) and does nothing out of sample, which is itself the signature of a bias rather than of under-dispersed noise. Rejecting deep twilights is actively harmful: it removes exactly the large-residual events that were inflating the variance, so the filter becomes more confident and less accurate at once, with lockbox coverage falling to zero. Every row retains a lockbox latitude bias of at least 413 km. None of these is a fix.

IX. Why the Fix Could Not Have Worked

Section VIII reports a bias state that improves calibration and not accuracy, and I wrote that up as a fix that failed. That is the right description of the measurement and the wrong diagnosis. The bias state could not have worked, and the reason is structural rather than empirical.

A. The bias is confounded with position by construction

A day of twilight data gives exactly two numbers, a dawn time and a dusk time. Position has exactly two unknowns. The measurement Jacobian $J = \partial(t_{\text{dawn}}, t_{\text{dusk}})/\partial(\varphi, \lambda)$ is therefore square, and generically invertible away from the equinox. Any additional parameter that enters through those same two numbers lies in the range of J: it is a linear combination of a latitude shift and a longitude shift, and no quantity of twilight data can separate it from one. This is not a weak-identifiability statement that more data would fix. It is exact.

TABLE VIII

POSITION DISPLACEMENT THAT IS OBSERVATIONALLY IDENTICAL TO ONE MINUTE OF THRESHOLD BIAS, MEDIAN BY HALF-MONTH, IN km. SIGNED, SO THE ANTISYMMETRIC COLUMN CHANGES SIGN AT THE EQUINOX WHERE $\partial(\text{DAY LENGTH})/\partial\varphi$ PASSES THROUGH ZERO. κ IS THE CONDITION NUMBER OF J.

Month	n	κ	common mode		antisym.
			λ	φ	φ
2008-02	23	3.5	-23.7	$+0.1$	$+96.8$
2008-03	30	14.1	-25.6	$+0.1$	$+245.7$
2008-04	28	4.4	-25.5	-0.1	-121.4
2008-05	21	2.4	-25.3	-0.0	-66.0
2008-06	5	1.8	-24.7	-0.0	-48.9

TABLE IX

TUNING-WINDOW MARGINAL LIKELIHOOD OVER A THREE-DIMENSIONAL BIAS FAMILY, THREE SEEDS AT 20,000 PARTICLES. UPPER BLOCK: THE ONE-DIMENSIONAL SWEEP OF TABLE VI RE-RUN THROUGH THE GENERAL CODE. LOWER BLOCK: THE FIVE CONFIGURATIONS THE EVIDENCE CRITERION ACTUALLY PREFERS.

σ_a	σ_c	σ_g	$\log Z_{\text{tune}}$	Tune	cov_{T}	Lock	cov_{L}
0	0	0	-1176.2	273.8	26%	365.9	20%
1	0	0	-1087.7	228.4	44%	594.3	0%
2	0	0	-1069.5	304.5	34%	457.6	27%
3	0	0	-1060.0	324.9	30%	544.0	13%
5	0	0	-1061.8	309.6	32%	414.8	20%
8	0	0	-1070.3	375.9	19%	564.4	15%
1	4	1	$\mathbf{-338.0}$	1165.5	11%	864.7	37%
2	4	1	-340.6	1087.8	12%	573.0	32%
3	4	1	-345.9	1069.6	11%	700.1	37%
0	4	1	-347.6	1548.7	7%	884.7	35%
1	4	2	-348.4	1174.3	12%	870.7	35%

The only free question is the exchange rate, which follows from solving $J\,\delta = e$ for the position displacement δ that reproduces a given observation perturbation e. A common-mode bias is $e = (+1, +1)$ minutes and an antisymmetric bias is $e = (+1, -1)$. Table VIII evaluates this at the interpolated Argos positions, so the rates are physical.

Over the record, one minute of common-mode bias is indistinguishable from $25.4\,\text{km}$ of longitude, and one minute of antisymmetric bias from $112.1\,\text{km}$ of latitude, rising in magnitude to $390\,\text{km}$ per minute in March. The $+34$ minute antisymmetric residual of Table V is nominally $3800\,\text{km}$ of latitude; the linearisation is long past valid at that size, so the usable content is the direction and not the number. At this exchange rate a biased threshold and a displaced animal are the same observation.

B. What happens when the selection rule meets a degenerate direction

Section VIII selects σ_b by tuning-window marginal likelihood and never touches the lockbox, and I treated that hygiene as sufficient. It is not. I widened the state to three bias dimensions — antisymmetric b_a, common mode b_c, and a depth coefficient g entering as $g\log(1+z)$ antisymmetrically — and ran the identical selection rule over 96 combinations of their random-walk scales, three seeds each.

Table IX is the result. Every configuration the evidence criterion prefers switches on the common-mode dimension, the one Table VIII shows is confounded with longitude at $25\,\text{km}$ per minute. The selected configuration scores $1165\,\text{km}$ on the tuning window against $274\,\text{km}$ for no bias state at all, a factor of 4.3 worse on the very window the criterion was computed on. Across all 96 configurations the rank correlation between $\log Z_{\text{tune}}$ and tuning-window error is $\rho = +0.76$ ($p = 4.8 \times 10^{-19}$), and between $\log Z_{\text{tune}}$ and tuning-window coverage it is $\rho = -0.57$. Higher evidence reliably means worse.

The mechanism is not subtle once stated. Marginal likelihood rewards explaining the observations. A degenerate direction explains observations perfectly, by absorbing whatever residual is left, and pays nothing for it because the data cannot contradict it. So evidence is maximised exactly where the model is least identifiable.

C. The filter doing it, in one number

Running the selected configuration and correlating the estimated common-mode bias against the signed longitude error gives Spearman $\rho = +0.94$. The filter is not measuring a property of the sensor. It is relabelling its own longitude error as a threshold offset, and scoring itself well for the relabelling.

D. Consequences

Three, in increasing order of how much they cost me.

The one-parameter fix of Section VIII is not a fix that failed, it is a parameter that was never estimable from twilight data. Its coverage gain is real but it is the honest widening of an interval around an unidentifiable quantity, not an improvement in knowledge.

Selection hygiene is necessary and not sufficient. A rule that never sees held-out data can still be the wrong rule, and mine was: it was safe in Table VI only because the family was one-dimensional and small. Widen the family and the same rule walks straight into the degenerate direction. I would now pair any evidence-based selection with an identifiability check on the parameters being selected over.

What breaks the degeneracy is information that does not arrive through the twilight pair. That is exactly what the SST and seafloor terms are, and it explains why they are worth more here than any filter, smoother or likelihood choice. The productive direction for this record was never a better estimator. It was a third independent measurement.

X. SMOOTHING: VARIANCE VERSUS BIAS

Section IX closes off the measurement-model repairs: any extra parameter that enters through a dawn/dusk pair is already in the span of (φ, λ). Smoothing is the one remaining idea it does not touch, because it adds no parameter at all. It only insists that the estimate at time t use the observations after t as well as those before.

The prediction is worth stating before the numbers, because it is the whole point. A smoother removes variance and lag. It

cannot remove a bias in the measurement model: if every twilight reports the same distortion, propagating that information backwards as well as forwards reports the distortion twice. So a large gain is expected on the tuning window, where the Argos fixes are dense and the residual error is mostly spread, and little or none on the lockbox, where Section IX says the error is structural.

A. Two smoothers, and three things checked before trusting either

I implement both a forward-filtering backward-simulation smoother (FFBS) and the generalised two-filter smoother. They approximate the same marginals by different routes, so agreement between them is evidence and disagreement is a bug.

The forward pass is the configuration of Section VIII: hard seafloor set, SST aiding, $12\,000$ particles. To store its full state cloud I patch the source of the published filter rather than retyping its loop, so the two cannot drift apart; the resulting track is bit-identical to the published one (maximum absolute difference 0). The transition density used backwards is checked against $150\,000$ simulated draws of the forward propagation and agrees to 3.6×10^{-15}.

For the two-filter smoother the artificial prior on velocity is taken to be the stationary law of the Ornstein–Uhlenbeck velocity and the backward proposal its exact time reversal, which makes the importance ratio $\gamma_t(x_t) f(x_{t+1} \mid x_t) / [\gamma_{t+1}(x_{t+1}) q(x_t \mid x_{t+1})]$ identically one. The code measures that ratio rather than trusting the algebra: the largest absolute log ratio over all 241 steps is 1.3×10^{-12}. Fusion is done between the particle clouds, not between Gaussian summaries of them.

B. A large gain on the tuning window, none on the lockbox

Table X gives the result over five seeds. On the tuning window the smoother cuts median error from 119.0 ± 1.4 to $68.6 \pm 4.4\,\mathrm{km}$, a gap of $50.3\,\mathrm{km}$ at 10.9 standard errors, and the geometric-median readout reaches $56.8 \pm 3.8\,\mathrm{km}$. This is the largest single improvement in the study, and it costs no new parameter and no new data.

On the lockbox the same smoother moves the error from 404.2 ± 2.9 to $388.2 \pm 22.8\,\mathrm{km}$, a gap of 0.7 standard errors, which is not a difference. The signed error is almost unchanged: $378\,\mathrm{km}$ too far south becomes $368\,\mathrm{km}$ too far south. The two smoothers also agree with each other to about $1\,\mathrm{km}$ on both windows despite sharing no backward machinery, which is the strongest single argument that the number is real.

Three controls support the reading. The answer is flat in the smoothing budget (66.5–$67.3\,\mathrm{km}$ for K from 100 to 1600 trajectories), so it is not a compute artefact. None of the $96\,400$ smoothed states leaves the hard seafloor set, so admissibility survives smoothing by construction rather than by luck. And the posterior spread falls to 0.65 of the filter's while 95% coverage on the tuning window *rises* from 92.2% to 93.8%: the intervals get narrower and more honest at the same time. On the lockbox coverage is 37.5% before and after, which is the signature of an interval centred in the wrong place rather than one that is too narrow.

TABLE X
FILTER AGAINST SMOOTHER, FIVE SEEDS, MEDIAN GREAT-CIRCLE ERROR IN km, MEAN ± STANDARD ERROR. THE LAST COLUMN IS THE MEDIAN SIGNED NORTH ERROR ON THE LOCKBOX, WHICH IS WHAT SEPARATES VARIANCE FROM BIAS. THE TWO-FILTER ROW IS A SINGLE-SEED CROSS-CHECK ON THE SAME FORWARD PASS.

Estimate	Tune	Lockbox	Δ_N
Filter, mean	119.0 ± 1.4	404.2 ± 2.9	-378
FFBS, mean	68.6 ± 4.4	388.2 ± 22.8	-368
FFBS, geom. median	56.8 ± 3.8	412.7 ± 18.3	-380
Two-filter, mean	70.3	357.3	—

One stated motivation does not survive. The reason usually given for smoothing here is lag, and there is very little lag to remove: shifting the filter's track in time buys only $1.6\,\mathrm{km}$, at a best shift of one day. The improvement comes from using future observations to sharpen each marginal.

C. A reason I had skipped this, and why it was wrong

An earlier step in the notebook rejected backward simulation on an explicit calculation: the position process noise over a median step is about $2\,\mathrm{km}$ while the posterior cloud is about $125\,\mathrm{km}$ across, so only ~ 3.5 of $12\,000$ particles lie within one standard deviation of a backward target. That arithmetic is correct and I reproduce it: the median effective number of ancestors per target is 2.9.

The conclusion drawn from it was not. Two claims were conflated. That the backward weights *underflow* is a statement about floating point, and it disappears once the weights are formed in log space with the row maximum subtracted. That the sampler *collapses* does not follow at all, because each trajectory draws its own ancestor independently; a concentrated weight vector pins one trajectory rather than collapsing the ensemble. Measured directly, 306 of 400 smoothed trajectories remain distinct at the start of the record, and Table X shows the ensemble is better calibrated than the filter it came from. Rejecting the method on that argument cost roughly $50\,\mathrm{km}$ of tuning-window accuracy.

XI. TEMPERATURE: TWO ACCIDENTAL CORRECTIONS

Section IX closes the twilight channel and Section X shows that a smoother removes variance but not bias. The one remaining place to look for independent information is the thermometer. The tag reports a surface temperature with each dawn, and the pipeline compares it to the NOAA OISST quarter-degree daily analysis through a Gaussian likelihood with one fitted bias and one fitted standard deviation, thinned to every third day to respect the analysis error correlation.

Two files in the same archive had never been opened: a binned depth–temperature profile roughly every two hours, and a mixed-layer summary. Neither supplies a second position fix. What they supply is a per-day statement of how far the one temperature reading can be trusted, and using them turned up a defect rather than an improvement.

A. The land mask was deleting the true position

Evaluating the residual at the *true* Argos position removes the position error and leaves the observation model exposed on its own. Done that way, the first thing visible is not a noise-model problem at all. On 19 of the 95 dawn events that carry both a temperature reading and a defensible truth—one fifth of them, and 14 of the 18 that fall in the lockbox—OISST returns no value at the animal's own Argos fix. Seventeen of those nineteen are water according to the same bathymetry the seafloor constraint already uses, at a median depth of 103 m. They lie in the Gulf of California, which is narrower than the cell the analysis is built on.

The likelihood mapped a missing reference to $-\infty$, reasoning that the animal cannot be on land. A gap in OISST does not mean land; it means land to a quarter-degree product. Averaged over sampled positions on days when the term fires, it was assigning zero probability to 35.3% of the map, the true position included. Nothing is lost by treating a missing reference as uninformative instead: the hard seafloor set of Section VIII already rejects land exactly, at far finer resolution, and it is the term that is supposed to be doing that job.

B. The tag measures its own reliability

The published term uses one standard deviation for all 137 days. Measured against the truth on the tuning window the residual spread is 1.11 °C where the filter assumes 0.69 °C, so the temperature term is overconfident by a factor of 1.6. More useful than the average is that the tag predicts its own error. Sorted into quartiles of the tag's own mixed-layer depth, the residual spread rises monotonically through 0.54, 0.75, 1.07 and 1.69 °C; a variance model linear in mixed-layer depth beats a constant one by $\Delta\mathrm{AIC} = -19.9$ on the tuning window. Near-surface stratification does about as well and is uncorrelated with calendar time ($r = -0.04$), so this is water-column physics and not a drifting sensor. A competing hypothesis died cheaply: the animal reached the surface on every single day the term fires, so a story about the tag never getting shallow enough explains nothing.

The same geometry that makes a twilight bias unidentifiable applies here. The meridional OISST gradient along this track is 1.07 °C per degree of latitude, so one degree of free thermometer bias buys 103 km of free latitude. A latent temperature-bias state is therefore the Section IX trap in a different channel. A bias regressed on an *observed* covariate adds no latent freedom and is the only safe form, and it is the only form tested below.

C. Repairing it rotates the error instead of shrinking it

Table XI runs four versions of the temperature term through the same forward filter. Against the published term, treating missing cells as uninformative and taking the per-day standard deviation from the mixed-layer depth improves the tuning-window mean by 23 km, the RMS by 29 km and the 75th percentile by 82 km, each past ten standard errors—while making the median 20 km *worse*. Both statements are true, and which one is quoted is a decision about whether a typical day or a bad day matters more. This paper has quoted the median almost everywhere, and the median is precisely the statistic that cannot see a defect which ruins one day in five.

TABLE XI
FOUR VERSIONS OF THE TEMPERATURE TERM THROUGH THE SAME FORWARD FILTER, FIVE SEEDS, GREAT-CIRCLE ERROR IN km. STANDARD ERRORS ARE 0.5–3.8 km ON THE TUNING COLUMNS AND 1.7–10.6 km ON THE LOCKBOX MEAN. P75 IS THE 75TH PERCENTILE.

Temperature term	Med.	Mean	P75	Lock
As published	118.5	166.9	273.1	446.5
Missing cell uninformative	129.1	145.7	176.8	452.1
+ per-day σ	138.8	143.8	191.0	450.2
+ per-day bias	161.3	164.2	228.3	481.3

The signed error says why the median moved the wrong way. The median tuning-window offset goes from 33 km south and 70 km west under the published term to 97 km south and 13 km west under the fully repaired one, while its magnitude stays between 70 and 98 km. The error does not shrink; it rotates from south-west to very nearly due south. Two defects had been cancelling. Deleting every coastal cell acted as an offshore repulsion that nobody wrote down, worth about 25 km of eastward error. The unmodelled 0.58 °C cold offset acted as a northward pull: at 1.07 °C per degree of latitude it should be worth 60 km, and the observed loss of northward pull when it is modelled away is 64 km. Repairing the measurement model honestly removes both accidental corrections and exposes the southward twilight bias that Section IX says cannot be removed at all.

One further negative is worth recording. The per-day *bias* term is preferred by AIC on the observation residual and is worse than the per-day standard deviation alone on every filter statistic in Table XI. Fitting the residual better is not the same thing as locating the animal better, and an information criterion computed on the residual cannot distinguish them.

D. What the repair is worth once you smooth

Run through the smoother of Section X instead of the filter, the repair almost vanishes: tuning-window mean error moves from 94.4 ± 3.7 to 92.5 ± 1.2 km, half a standard error, with the 75th percentile improving by 14.7 ± 6.6 km and the lockbox mean going the wrong way by 20.2 ± 9.5 km. The filter needed the repair because it has only the past to work with, and a day on which the truth is deleted is unrecoverable. A smoother that can see both directions repairs those days by itself.

The conclusion for the temperature channel is therefore negative in the useful sense. It held no untapped accuracy. What it held was an unflagged defect that was making the filter wrong in a flattering direction, and the tag's own depth–temperature profiles were enough to find it without a single new data source. The repair should be adopted anyway, on the grounds that a likelihood must not assign probability zero to a place the animal demonstrably was, and not on the grounds that it improves a score.

XII. Process Noise and Habitat Priors

The last item on the list was the motion model and a weak preference for plausible habitat. I expected the smallest return of the three and got the largest lockbox improvement in the project, attached to a result that does not mean what it appears to mean.

A. One day of data cannot find this animal

Before adding a prior I wanted a number for how much the measurement model knows on its own. Both checks below use the 176 of 241 events whose true position can be defended by interpolation between Argos fixes less than four days apart. The hard feasible set contains that position on 98.3% of them, so the constraint is not what is hurting.

The first check evaluates the filter's own measurement likelihood—the same code, asserted line for line against the body of the filter—at the true position and at the filter's answer. The answer scores higher than the truth on 51 to 70% of events depending on the readout and the window, and the median log-likelihood gap is 0.01 to 0.22. Being 430 km from the animal costs about a tenth of a log unit.

The second check deletes the filter. On a 351×351 grid at 0.1° spacing, 123,201 cells masked to the feasible set, I evaluated one solar day of twilight and temperature data at every grid point, for each of the 104 days with defensible truth. The best point on that map lies 372.2 km from the animal on the tuning window and 533.2 km in the lockbox. The truth sits in the top 1.6% and 11.1% of the map respectively, and the set within two log units of the maximum covers 3.7 to 4.7% of the map, several hundred kilometres across.

One day of data therefore rules out most of the ocean and locates nothing inside what is left. Every kilometre of accuracy in Table II comes from the motion model stitching a hundred vague days together. A prior is not competing with the data for influence; there is very little to compete with.

B. The motion model was fitted on the month the animal was migrating

σ_v was set from net daily displacement over the calibration window. Splitting the tuning-window Argos fixes at the end of that window shows what that choice bought: median displacement is 68.1 km per root-day over the 12 calibration pairs and 16.2 over the remaining 34, a factor of 4.2. The calibration window is exactly the stretch in which the animal ran south from Monterey. Over the whole tuning window the figure is 23.8, and in the lockbox, which was not consulted, 28.5.

A prior-predictive check makes the consequence concrete. Simulating the published motion model freely across the 47 real Argos gaps in the tuning window gives a median displacement of 78.4 km against 32.4 km observed: a ratio of 2.42 and a two-sample Kolmogorov–Smirnov statistic of 0.349 ($p < 10^{-4}$). The scale that reproduces the observed median is 0.40; at 0.5 the ratio is 1.22 and the test no longer rejects ($p = 0.083$). The published filter swims about two and a half times too hard.

TABLE XII
Process-noise scale swept through the forward filter at 12,000 particles with the hard feasible set, three seeds, great-circle error in km. Standard errors are 0.5–16.4 km. The scale is chosen on the tuning columns alone.

σ_v	Med.	Mean	P75	Lock med.
$\times 0.25$	286.7	372.8	590.3	478.5
$\times 0.5$	99.4	140.1	215.1	312.5
$\times 1$ (published)	117.7	167.0	270.2	411.8
$\times 2$	179.3	222.1	347.2	588.0

TABLE XIII
Same weight ($\lambda = 1$) and the same machinery; only the contents of the depth histogram change. Five seeds, 12,000 particles, great-circle error in km. Standard errors are 1.0–38.3 km on the tuning columns and 0.9–16.2 km on the lockbox median.

Depth prior	Med.	Mean	Lock med.	ESS
None (published)	118.5	166.9	433.2	7174
Fitted on tuning window	83.8	111.1	290.6	6272
Preference inverted	91.5	203.8	240.1	4242
Shuffled, draw 0	415.5	435.4	585.7	6453
Shuffled, draw 1	79.1	170.4	290.0	4969
Shuffled, draw 2	415.5	436.6	586.1	6471
Shuffled, draw 3	83.2	176.7	255.5	4867
Shuffled, draw 4	74.5	262.7	271.6	4908

Table XII sweeps the scale through the forward filter. Halving σ_v improves all four tuning-window statistics at once, so it is selected without the lockbox being consulted, and it transfers: lockbox median error falls from 411.8 to 312.5 km. This is the same failure as the light threshold in Section VII-C, a parameter fitted on a window whose conditions do not hold for the rest of the deployment, in a parameter I had not thought to check.

C. A weak habitat prior, and the control that ruins it

The prior is a histogram. I read the seafloor depth under each of the 91 tuning-window Argos fixes—quartiles 100, 331, 789, 1379 and 3165 m, with 62.6% of them over water shallower than 1000 m—and binned $\log_{10}$ depth into 18 bins of width 0.25, of which 9 are occupied. Laplace smoothing, peak normalised to zero, added to the log-likelihood with weight λ. Eighteen numbers, all of them from the tuning window, and nothing in any of them about position.

At $\lambda = 1$ it improves all eight statistics at once (Table XIII): tuning median 118.5 to 83.8 and lockbox median 433.2 to 290.6 km. That is the largest single lockbox gain anywhere in this study, out of eighteen numbers.

It is also wrong. Keeping the weight and the machinery and changing only the contents of the histogram, I inverted the preference so that the prior pushes the animal towards exactly the depths it avoids, and separately shuffled the bin heights at random five times. The inverted prior beats the correct one on the lockbox median, 240.1 against 290.6 km. Three of the five shuffled priors also beat the published filter by more than 140 km. Whatever the depth term is supplying, it is not habitat knowledge.

TABLE XIV
COMBINING THE TWO INTERVENTIONS. FIVE SEEDS, 12,000 PARTICLES, GREAT-CIRCLE ERROR IN km; STANDARD ERRORS ARE 1.0–7.7 km. THE LOWER PANEL GIVES ERROR PERCENTILES POOLED OVER THREE SEEDS, AND SHOWS WHAT THE BEST LOCKBOX ROW COSTS ON THE WINDOW THAT IS ALLOWED TO SELECT IT.

Configuration	Med.	Mean	Lock med.	Lock mean
Neither (published)	118.5	166.9	433.2	446.5
Depth prior only	83.8	111.1	290.6	287.7
σ_v halved only	98.5	138.0	319.1	319.0
Both	60.6	205.1	165.0	158.8
Error percentiles, three seeds pooled				
	P50	P75	P90	Max
Published, tuning	120	268	329	450
Both, tuning	61	340	566	903
Published, lockbox	409	581	586	702
Both, lockbox	165	197	240	271

D. What the depth term is actually doing

The mechanism is confinement. For each of the seven depth priors I measured the fraction of the ocean it leaves at full weight—its peak-normalised value summed over water cells, divided by the number of water cells—and the position of its weighted centre of mass. Effective area ranges from 7.2% to 63.3%, and across the seven priors its correlation with lockbox median error is $+0.951$. The correlation with how close the prior's centre of mass sits to the animal is $+0.740$, but area and distance are themselves correlated at $+0.880$ across these seven; holding distance fixed, the area correlation is $+0.940$, and holding area fixed the distance correlation changes sign to -0.667. Every one of these priors, including the two that help most, has its centre of mass more than $1000\,\mathrm{km}$ from the animal's mean lockbox position. They are not pointing at the animal. They are narrow, and against a likelihood as flat as the one measured above, narrow is enough.

The same argument predicts that the two interventions should combine, since tightening σ_v confines in time and the depth term confines in space. Table XIV shows that they do, and almost exactly additively: together they give a lockbox median of 165.0 ± 1.2 and a lockbox mean of $158.8 \pm 3.7\,\mathrm{km}$, against 433.2 and 446.5 for the published filter. That is the best lockbox accuracy in the project by a factor of 2.6.

E. What I adopt, and what I only report

Halving σ_v is adopted. It is chosen by the tuning window on all four statistics, it has a prior-predictive justification that does not reference error at all, and it transfers.

The depth prior is reported and not adopted, because its own control beats it. The combination is reported and not adopted for a different reason: on the tuning window it has a 99th percentile of $872\,\mathrm{km}$ and a worst day of $903\,\mathrm{km}$, and its tuning mean and RMS are worse than the published filter's. It is excellent four days in five and badly wrong on the fifth. Under this study's own rule—select on the tuning window, score the lockbox once—it is rejected. The rule that stops me adopting it is the same rule that would have stopped me ever finding it, which is a fair statement of what held-out evaluation costs as well as what it buys.

If the measurement model is this flat, anything that shrinks the support will look like skill. The only way I found to tell confinement from information was to build a deliberately wrong version of the same term and check whether it worked too.

XIII. DOES ANY OF THIS TRANSFER?

Everything above is one animal, which is the first thing anyone should attack. The archive these data come from holds 70 deployments, and seven of those animals carried a PAT and a SPOT tag at the same time. Those seven are the only animals in the database that can be scored the way 07_05 was scored, because they are the only ones with an independent position while the light tag was running.

I ran the frozen configuration on all seven. The threshold h_0, the attenuation coefficient β, the twilight noise model, σ_v and the halving adopted in Section XII are exactly the values 07_05's tuning window produced. Three deployment facts change per animal—release time, release location, and the animal's own archival depth record—and nothing else. For the six new animals there is no tuning window at all: every day is held out, and 07_05 stays in the table as a control whose numbers must not move.

A. Two defects that one animal could not reveal

The first run of this experiment produced a table I could not defend, and chasing why exposed two bugs. Both were silent: no exception, no warning.

The seafloor constraint was the first. `ZWIN`, the per-twilight maximum dive depth that makes a position admissible, was built once as a dictionary keyed by 07_05's timestamps. On any other animal every lookup missed, the depth returned not-a-number, the test was skipped, and the feasible set degraded silently to ocean-only. Measured directly, the lookup succeeded on 95% of 07_05's twilights and on 0% of every other animal's. The six new animals were being scored under a weaker constraint than the animal they were being compared against, and I would have reported the gap as a generalisation result.

The release epoch was the second and the worse of the two. The filter starts its clock at a module-level constant holding 07_05's deployment date. 07_03's record ends two months *before* that date, so every time increment came out negative, the propagation block never executed, and the filter sat frozen on its release prior for all 232 of its twilights. A stationary estimate near a release site scored $85\,\mathrm{km}$, the lowest number anywhere in this study, and halving the process noise changed it by exactly zero—identical to fifteen significant figures. That zero is the only reason I found the bug. The other five animals each received one spurious first step of 135 to 600 days; 09_15's 600-day step alone accounted for a $1187\,\mathrm{km}$ median and a $483\,\mathrm{km}$ seed spread.

Both repairs have the same shape as the release prior: swap a per-animal deployment fact, touch no tuned parameter. The

TABLE XV
THE FROZEN CONFIGURATION ON SEVEN DOUBLE-TAGGED ANIMALS, 8,000 PARTICLES, THREE SEEDS, MEDIAN GREAT-CIRCLE ERROR IN km AGAINST EACH ANIMAL'S OWN ARGOS FIXES. ONLY 07_05 WAS EVER TUNED ON; FOR THE OTHER SIX EVERY DAY IS HELD OUT. TWI. IS SCORABLE TWILIGHTS, FIX IS MATCHED ARGOS COMPARISONS.

Animal	Twi.	Fix	Published	$\sigma_v/2$	Change
07_05	241	115	364.4 ± 5.6	333.4 ± 13.4	-8.5%
07_03*	232	140	526.3 ± 28.9	477.0 ± 3.6	-9.4%
08_01	169	47	166.3 ± 0.8	130.4 ± 2.3	-21.6%
08_02	225	102	162.7 ± 1.0	135.9 ± 1.5	-16.4%
08_04	261	148	234.4 ± 0.9	180.9 ± 1.3	-22.8%
08_09	261	139	308.8 ± 1.3	233.3 ± 1.2	-24.5%
09_15	64	18	212.8 ± 2.1	143.8 ± 1.5	-32.4%

*No archival record on this tag, so the depth test is off for this animal and reported as such rather than hidden.

regression test is that 07_05 rebuilt through the new per-animal path reproduces the original inline result exactly—241 events against 241, maximum absolute difference in observed twilight hour 0, and the same detachment timestamp—and that its scores are unchanged to every digit in Table XV.

B. What transfers

Halving the process noise helps all seven animals, by 8.5 to 32.4% in median error with a median of 21.6%, and on six of the seven the gap is many times the seed-to-seed spread. This is the one intervention in the study that was selected on tuning data alone and then held everywhere it was tried. It is also the least clever: the process-noise scale was fitted on the single month 07_05 was migrating, and the finding is simply that a motion model fitted on an animal's fastest month is too loose for every animal's typical month.

Two animals do not deserve three seeds. 07_03 has no archival record and the largest seed spread in the table; 09_15 contributes only 64 scorable twilights and 18 matched fixes. Both were re-run with ten seeds before being quoted. 07_03 moves from 526.3 ± 28.9 to $516.2 \pm 30.5\,\mathrm{km}$ published and sits at $477.0 \pm 2.4\,\mathrm{km}$ halved; 09_15 moves from 212.8 ± 2.1 to 212.6 ± 1.8 and from 143.8 ± 1.5 to 143.9 ± 2.0. Both conclusions hold.

What this does not show is that the method is competitive. Six of seven medians sit between 130 and 330 km, well above the 118 km the proprietary smoother reaches on 07_05. The transferable result is about the process model, not about the estimator.

XIV. A Reference Field That Covered One Animal

Section XIII found two constants frozen to 07_05. This is the third, and it was in the data rather than in the code.

The temperature likelihood needs a gridded reference field. The one fetched for Section XI was requested for exactly the window needed at the time—2008-02-01 to 2008-06-30, which is 07_05's deployment and nothing else. Every line built on top of it is correct; it simply had nothing to read for any other animal. Of the 1052 dawn events in the cohort that carry a tag temperature, 149 fall inside that cube, and three of the seven animals have no reference data at any point in their record. The seven-animal table of Section XIII is therefore scored with the temperature channel switched off. That is stated there, so the table is not wrong, but it measures the filter with the one observable that carries latitude withheld.

Re-requesting the same dataset over the same spatial box for 2007-07 to 2010-03 raises coverage to $1052/1052$. The old and the new cube agree to better than 10^{-9} wherever they overlap and their missing-data masks agree exactly, so no number already reported moves.

A. A missing reference value is ignorance, not impossibility

The shipped likelihood maps a missing reference to $-\infty$: the animal was in the water, so a land cell is impossible. Inside a cube that covered the animal this is nearly harmless. Across the cohort it is wrong twice over. A missing value means either that the 0.25° land mask removed the cell—and the feasible set already excludes land, so scoring it here counts the same evidence a second time—or that the particle is off the cube, which is ignorance rather than impossibility. Roughly half of a plausible cloud lands on such a cell: sampling a 3° Gaussian around each release point gives missing-reference fractions of 0.49 to 0.57 across the seven animals. Declaring that half impossible never trips the collapse counter, because the filter has a fallback, but it empties the cloud: at the worst step the effective sample size falls to 4 particles out of 8,000 on 07_05 and 6 on 08_09, against 48 and 189 under the corrected rule. The honest increment for a missing reference is zero.

B. A per-tag offset cannot be estimated without the answer

Each tag's temperature sensor has its own offset, and the one in use was calibrated on 07_05. The only way to estimate a per-tag offset without reading an Argos fix is to compare the tag against the reference at the known release position for the first few days of the record. That estimator recovers the offset obtained with access to the truth with a mean absolute error of $0.85\,^\circ\mathrm{C}$ and a correlation of 0.23, against $0.26\,^\circ\mathrm{C}$ for simply reusing the frozen constant. Scoring it confirms the prediction: it helps 07_03 and 08_01 and costs 08_04 and 09_15 19 and 30% respectively. The oracle version, fitted on each animal's own Argos fixes and therefore not a result I am entitled to quote, buys under 6% beyond the frozen constant on five of the seven animals and 11 to 13% on the other two. Per-tag temperature calibration is not where the remaining error is.

C. What temperature is worth across the cohort

Table XVI is the result. All seven animals improve in median, mean and RMS error. Across the six animals that were never tuned on, the median improvement is 18.4%; including 07_05 it is 20.9%. 07_05 falls from 333 to 159 km and 09_15 from 144 to 72 km. The three animals most likely to be reporting noise—07_05 because it moved furthest, 08_04 because it barely moved, 09_15 because it has only 18 matched fixes—were re-run with ten seeds, and each reproduces its three-seed conclusion: 328.0 ± 15.1 to 148.5 ± 14.6, 179.2 ± 1.3 to 175.2 ± 1.4, and 142.5 ± 0.9 to 72.8 ± 4.1 km.

TABLE XVI

TEMPERATURE AIDING TRANSFERRED TO THE COHORT. FROZEN 07_05 CALIBRATION—SAME NOISE SCALE, SAME OFFSET, SAME THREE-DAY THINNING—WITH THE CORRECTED MISSING-DATA RULE AND NO OTHER CHANGE. 8,000 PARTICLES, THREE SEEDS, MEDIAN GREAT-CIRCLE ERROR IN km AGAINST EACH ANIMAL'S OWN ARGOS FIXES. THE NO-SST COLUMN IS THE $\sigma_v/2$ COLUMN OF TABLE XV. † MARKS THE ANIMAL THE CALIBRATION WAS FITTED ON; THE OTHER SIX ARE THE GENERALISATION EVIDENCE.

Animal	Fix	No SST	With SST	Change
07_05†	115	333.4 ± 13.4	159.1 ± 3.3	-52.3%
07_03	140	477.0 ± 3.6	401.5 ± 12.8	-15.8%
08_01	47	130.4 ± 2.3	80.6 ± 0.8	-38.1%
08_02	102	135.9 ± 1.5	107.6 ± 0.9	-20.9%
08_04	148	180.9 ± 1.3	176.2 ± 1.4	-2.6%
08_09	139	233.3 ± 1.2	199.9 ± 4.6	-14.3%
09_15	18	143.8 ± 1.5	72.2 ± 1.7	-49.8%

The weakest entry is 08_04 at -2.6%, about three times its seed spread and the only animal whose 75th percentile moves the wrong way (229 to 247 km). I do not count it as a win. The honest summary is that the observable which carries latitude was being withheld from six of seven animals by the extent of a downloaded file, and that restoring it is worth more than any change I made to the estimator. The offset behind the with-SST column is still 07_05's; Section XV asks whether it can be shared.

XV. SHARED CONSTANTS, AND A LOCKBOX TOO SMALL TO RULE

Section XIV applies constants fitted on 07_05 to six animals that never influenced them. That is a fair zero-shot test, but it is a weak claim about a *shared* parameter: one animal is setting a constant for six others. This section redoes the two constants that matter leave-one-animal-out, and re-opens the lockbox for the corrected temperature rule, because a corrected rule is a different filter from the one the lockbox was opened on.

A. Re-opening the lockbox, once

The lockbox is the last 30 days of 07_05's Argos record and was scored once, with the shipped missing-data rule in place (Section V). Section XIV showed that rule was wrong. I score the corrected rule against the lockbox one further time, with ten seeds, and then not again. Nothing in this paper selects on the result.

Table XVII gives both windows. On the tuning window, with 91 fixes, the median falls from 159.8 ± 3.0 to 144.6 ± 6.5 km, a change of 2.3 times the larger seed spread. On the lockbox, with 24 fixes, it falls from 281.1 ± 71.9 to 212.1 ± 81.5 km: nominally -24.6%, but only 0.8 times the larger spread. The improvement is measurable on the window where I can measure it and unresolvable on the window that was supposed to arbitrate it. Twenty-four fixes are too few to settle a change of this size, which is a defect in the held-out set I designed rather than in the fix. A held-out set should be sized by the number of scorable events it lands on, not by the calendar span it covers.

TABLE XVII

THE LOCKBOX RE-SCORED UNDER THE CORRECTED MISSING-DATA RULE. 07_05, 8,000 PARTICLES, TEN SEEDS, MEDIAN GREAT-CIRCLE ERROR IN km, MEAN ± STANDARD DEVIATION OVER SEEDS. THE TWO ARMS DIFFER ONLY IN WHAT A MISSING REFERENCE TEMPERATURE IS ALLOWED TO MEAN.

Window	Rule	Fix	Median	Change
Tuning	shipped	91	159.8 ± 3.0	
	corrected	91	144.6 ± 6.5	$2.3\times$ sd
Lockbox	shipped	24	281.1 ± 71.9	
	corrected	24	212.1 ± 81.5	$0.8\times$ sd

TABLE XVIII

TEMPERATURE OFFSET, FROZEN AGAINST POOLED AGAINST ORACLE. MEDIAN GREAT-CIRCLE ERROR IN km, THREE SEEDS, EVERYTHING ELSE HELD AT THE FROZEN 07_05 CALIBRATION. FROZEN IS 07_05'S OFFSET APPLIED TO ALL SEVEN; RELEASE IS EACH TAG'S OWN RELEASE-WINDOW ESTIMATE; POOLED IS THE MEAN OF THE OTHER SIX ANIMALS' TRUTH-FITTED OFFSETS, WHICH NEVER SEES THE HELD-OUT ANIMAL. * IS FITTED ON THE ANIMAL'S OWN ARGOS FIXES AND IS AN UPPER BOUND, NOT A RESULT I AM ENTITLED TO QUOTE.

Animal	Frozen	Release	Pooled	Oracle*
07_05	159.1 ± 3.3	155.5 ± 6.2	141.8 ± 9.5	150.8 ± 2.7
07_03	401.5 ± 12.8	390.7 ± 7.6	388.0 ± 14.8	400.3 ± 8.6
08_01	80.6 ± 0.8	67.1 ± 1.0	78.1 ± 0.7	77.1 ± 0.2
08_02	107.6 ± 0.9	115.3 ± 0.8	104.9 ± 1.6	109.6 ± 0.7
08_04	176.2 ± 1.4	215.9 ± 0.8	159.8 ± 1.0	152.9 ± 0.1
08_09	199.9 ± 4.6	207.8 ± 3.2	192.5 ± 3.1	177.3 ± 5.2
09_15	72.2 ± 1.7	100.3 ± 3.3	70.4 ± 3.6	72.0 ± 1.5
Mean	171.0	178.9	162.2	162.8

B. A temperature offset that pools

For each animal I set the temperature offset to the mean of the other six animals' truth-fitted offsets. The held-out animal contributes nothing to its own constant, so unlike the oracle column of Table XVI this is a quantity I am entitled to quote.

Table XVIII gives the comparison. The pooled offset beats the frozen 07_05 offset on seven animals out of seven (one-sided sign test, $p = 0.008$). Per animal the gain is small—the median change is -3.3%—but the direction is unanimous, and the cohort mean of the per-animal medians falls from 171.0 to 162.2 km against 162.8 km for an oracle fitted on each animal's own fixes. Averaging six noisy estimates is as good as handing each animal its own answer, which is what one expects when the per-animal estimates are dominated by estimation noise rather than by real differences between tags. The per-tag release-window estimate, which uses no truth at all, is the worst of the three on average at 178.9 km; it is worse than the frozen offset on four of the seven animals, and on 08_04 it is worse than using no temperature at all.

C. Process noise does not pool, and the obstacle is the sampler

The same treatment applied to the process-noise scale fails, and the failure is the more useful of the two results.

TABLE XIX

PROCESS-NOISE SCALE SWEPT ACROSS THE COHORT, MEDIAN GREAT-CIRCLE ERROR IN km, THREE SEEDS, POOLED TEMPERATURE OFFSET THROUGHOUT. σ_v IS MULTIPLIED BY THE SCALE; 0.50 IS THE VALUE ADOPTED IN SECTION XII. SIX OF THE SEVEN ANIMALS IMPROVE MONOTONICALLY AS THE SCALE FALLS. 07_05, THE ANIMAL THAT CHOSE 0.50, IS THE EXCEPTION.

Animal	0.25	0.35	0.50	0.70	1.00
07_05	727.1	704.9	141.8	142.7	154.3
07_03	318.1	379.8	388.0	339.5	406.1
08_01	51.7	66.4	78.1	91.2	117.1
08_02	71.4	84.4	104.9	129.7	164.8
08_04	105.0	129.0	159.8	182.1	217.5
08_09	119.0	152.3	192.5	231.1	273.2
09_15	59.7	62.9	70.4	128.1	195.9
Mean	207.4	225.7	162.2	177.8	218.4

Table XIX sweeps the scale. Six of the seven animals improve monotonically as it falls—08_09 runs from 192.5 to 119.0 km between 0.50 and 0.25—while 07_05, the animal that set 0.50 in Section XII, collapses instead: 141.8 km at 0.50 and 704.9 km at 0.35. A leave-one-animal-out vote therefore reproduces the frozen 0.50 on six of the seven folds, and on the fold that holds 07_05 out the other six vote 0.25, where 07_05 scores 727.1 km against 141.8 km frozen.

A discontinuity of that shape is not a bias–variance trade-off, and the standard particle-filter health check does not see it. The effective sample size never falls below 4,027 of 8,000 and no step is flagged as collapsed or redrawn. The weights stay near-uniform precisely because every particle is equally far from the animal: the filter is not degenerate, it is confidently wrong. What does move the number is the particle budget. At scale 0.35, 07_05 scores 704.9 ± 11.1, 531.5 ± 80.8 and 233.7 ± 60.8 km at 8,000, 32,000 and 128,000 particles, and the trend has not flattened.

Sixteen times the budget removes two thirds of the gap and still does not reach the 141.8 km that 0.50 achieves with 8,000, so I cannot claim the tighter process noise wins, and the shipped configuration stays at 0.50. What I can say is that most of the cliff belongs to the sampler rather than to the animal, and that the adopted value should be read as the smallest process noise this sampler survives on the hardest animal rather than as an estimate of how a white shark moves. Every process-noise number in this paper is to that extent a statement about compute, and the honest next step is a better proposal—a regularised or auxiliary particle filter—before this grid is swept again.

XVI. DISCUSSION

A. What to remember

1) **Tuning-window numbers did not generalise.** The best tuning-window configuration is near the worst on the held-out window, and rank on one barely predicts rank on the other (Section VI-E).
2) **Twilight bias parameters are confounded with position.** Two observations per epoch cannot identify position and a threshold bias at once, so that whole family of fixes was unavailable from the start (Section IX).
3) **The largest gains came from data and reference handling, not from the estimator.** A reference field downloaded for one deployment window, and a likelihood that read a missing reference as an impossibility, cost more than every filter choice in the study combined (Section VI-A, Table I).
4) **The temperature offset pools across animals; the process-noise scale does not.** One constant set leave-one-animal-out matches an oracle fitted on each animal's own fixes; the other collapses on the animal that set it, for reasons that belong to the sampler (Section XV).
5) **The lockbox is powered for large effects only.** With 20 matched twilight events and 24 Argos fixes, seed spread alone is 80–90 km, so nothing below roughly 170 km clears twice that. It was enough to show that the headline was a tuning-window number, and it is not enough to adjudicate a 25% change (Section V).
6) **GPE3 is the only estimator that holds up on both windows.** It is the external benchmark here, it is ahead on all seven animals by factors of 1.8 to 7.6, and nothing I built closes the gap (Tables II and I, Fig. 4).

B. What survives

A Student-t twilight likelihood is worth about 100 km and is the largest single modelling win in this work — considerably more than the choice of filter, which is worth about 26 km. Smoothing beats filtering consistently for all three estimators, and a particle smoother is the single largest improvement in the study: 119.0 ± 1.4 to 68.6 ± 4.4 km on the tuning window at 10.9 standard errors, with the posterior narrowing to 0.65 of its former spread while coverage improves (Section X). Independent aiding is what breaks the equinox degeneracy: SST cuts March error from 522 to 109 km, because its information is orthogonal to day length rather than because it is more accurate. And once a hard constraint makes the posterior multimodal, the readout stops being innocent: the same particle cloud reads 364 km at the posterior mean and 452 km at the local MAP. Against four alternatives on the same clouds — marginal mode, weighted geometric median, highest-weight particle and a single weighted draw — the posterior mean is nonetheless the best choice on the tuning window, and none of the four beats it out of sample by more than its seed noise. One process-model parameter does survive on its own terms: σ_v was fitted on the calibration month, a prior-predictive check at the real Argos gaps rejects it at $p < 10^{-4}$, and halving it is chosen by the tuning window on all four statistics and cuts held-out median error from 412 to 313 km (Section XII). And one constant survives as a genuinely shared parameter: a temperature offset pooled over the other six animals beats the frozen single-animal offset on seven of seven, matches an oracle fitted on each animal's own fixes, and is chosen without ever seeing the animal it is scored on (Section XV).

C. What does not

The headline, as a measure of accuracy. The ablation ladder, which was built below the seed-noise floor. The raw-light recalibration, which improved the quantity it was fitted to and tripled the held-out error. The bias-state fix, as an accuracy claim, and in fact as any kind of estimate: Section IX shows the parameter was never identifiable from twilight data. And my own selection rule, as a sufficient safeguard — it is clean, and over a wider family it is actively anti-correlated with accuracy. And the temperature channel, as a reservoir of further accuracy: Section XI finds a real defect in it — the satellite analysis has no value at the animal's own position on a fifth of the days the term fires, and the likelihood called those places impossible — but repairing that defect rotates the error rather than reducing it, and buys nothing that survives smoothing. And the weak habitat prior, as habitat knowledge: it produces the largest single held-out gain in the study and is then beaten on that same window by a deliberately inverted copy of itself. Across seven such priors, how much ocean each one deletes explains the gain ($r = +0.951$), and where each one points explains none of it once area is held fixed (Section XII). And the halved process noise, as a statement about animal movement: six of the seven animals prefer a tighter scale than the one 07_05 chose, and 07_05's refusal to follow them is mostly sampler convergence rather than biology (Section XV).

D. Limitations

The diagnostic work is one animal, one deployment, and one ground-truth track; only the process-noise result of Section XIII, the temperature result of Section XIV and the pooled constants of Section XV have been tested on more. The lockbox contains 20 matched events for the twilight estimators and 24 Argos fixes for the particle filter, so lockbox medians carry wide intervals. As Section V quantifies, that is enough resolution to have shown the headline was a tuning-window number and not enough to adjudicate the 25% change of Section XV. I sized the lockbox by calendar span rather than by the number of scorable events it would land on, and that is the mistake. Finally, the particle budget is itself a limitation rather than a detail: at $8{,}000$ particles the filter is not converged at the process-noise scales most of the cohort prefers, so every process-noise conclusion is conditional on compute. The leave-one-animal-out protocol is also not perfectly clean for the process-noise vote: each training animal is scored with its own pooled offset, which includes a contribution from the held-out animal, so the vote carries a second-order dependence on the animal it is supposed to exclude. The offset comparison of Table XVIII does not, because each animal's constant is an average over the other six only. Nothing here should be read as a general ranking of nonlinear filters; it is a statement about what this data set can resolve, which is less than I originally claimed. The Argos error model for classes 0, A and B is assumed rather than known. GPE3 is a black box, so I can say that it generalises better but not why.

E. What I would do differently

Hold out a block of ground truth before writing any model code — a block at the end, scored once. Report seed variance on every stochastic result and refuse to interpret any difference smaller than it; two seeds would have killed most of the ladder. And treat calibration support as a first-class check: for every fitted sensor parameter, compare the range of its regressors in the calibration window against the range expected at deployment. A goodness-of-fit plot cannot show you an unidentifiable parameter; a support check can. I would also be slower to reject a method on a back-of-envelope argument: I skipped backward simulation because a correct calculation about weight concentration was attached to an incorrect conclusion about ensemble collapse, and that one unchecked inference cost about $50\,\mathrm{km}$ (Section X). Above all, before adding any parameter to a state vector, count observations per epoch against unknowns per epoch: that takes a minute and would have told me that the whole of Section VIII was estimating a quantity the data cannot see. I would also report more than one error statistic. Almost every number here is a median, and a median cannot see a defect that ruins one day in five: the repair in Section XI improves the mean by $23\,\mathrm{km}$ and the 75th percentile by $82\,\mathrm{km}$ while making the median $20\,\mathrm{km}$ worse. And I would open every file in the archive before modelling rather than after; the evidence that the temperature term was broken had been sitting in two files the pipeline never read. And for anything that improves a score by shrinking the posterior rather than by adding information, I would build a deliberately wrong version of the same term and check whether it works too; that one control is what separated Section XII's real result from its apparent one.

XVII. Conclusion

Light-level geolocation of these records is dominated not by the choice of nonlinear filter but by a biased measurement model that no filter in the family can detect from the data it is given, and by the handling of the reference data that filter is fed. Across seven animals, one frozen pipeline and no per-animal tuning, the changes that mattered were a process-noise scale that had been fitted on the wrong month, a temperature field that covered one deployment window out of seven, and a constant averaged over six animals instead of one. Together they take the cohort mean of per-animal medians from 282.2 to $162.2\,\mathrm{km}$, and none of it came from a better estimator. On the single animal I was allowed to tune on, my best configuration reaches $117\,\mathrm{km}$ and approaches the manufacturer's smoother. On 30 days I never looked at, it reaches $544\,\mathrm{km}$ against the smoother's $118\,\mathrm{km}$, and reports a $201\,\mathrm{km}$ standard deviation while being wrong by $532\,\mathrm{km}$.

The measurable contributions are therefore negative, and I think more useful than the positive claims they replace. Seed-to-seed spread of 29–$90\,\mathrm{km}$ invalidates an ablation ladder built from 1–$5\,\mathrm{km}$ steps. Tuning-window rank predicts held-out rank only weakly ($\rho = +0.30$), so the best-on-tuning variant ranks 54th of 75 out of sample. A recalibration that halved its own fitting residual tripled held-out error,

because the parameter it adjusted was never identifiable from the calibration window. And a principled one-parameter fix, chosen without ever touching the held-out data, improved calibration and not accuracy — and could not have improved accuracy, because a twilight pair is two numbers, position is two unknowns, and a threshold bias is therefore a linear combination of the two things it was meant to correct. The same marginal-likelihood rule, given a three-dimensional bias family, prefers the degenerate direction in every one of its top configurations and lands a factor of four worse on the window it was computed on.

A smoother — the one repair identifiability does not forbid — then halves tuning-window error while leaving held-out error statistically untouched, which is the cleanest confirmation available that the residual failure is bias and not variance.

The last two experiments make the same point from the other side. The process-noise scale was fitted on the only month of the record in which the animal was migrating; a prior-predictive check that never looks at error says it is 2.4 times too fast for the rest of the deployment, and halving it is selected on tuning data and carries to the held-out window. A weak seafloor-depth prior does better still, the largest single held-out gain anywhere here, and then loses to an inverted copy of itself, because what it supplies is not knowledge of where the animal is but a smaller region for it to be in. When one solar day of data places its own best estimate $372\,\mathrm{km}$ from the animal, almost anything that narrows the posterior will look like skill. None of this says habitat information is worthless; it says this particular prior earned its gain by shrinking the feasible area, so any habitat term has to be scored against a deliberately wrong copy of itself before it is credited with anything.

Extending the study from one animal to seven changes which of these claims survive. Two of the pipeline's defects were invisible on the tuned animal and obvious on the others, and the largest effect anywhere in this work turned out to be a reference field that had only ever been downloaded for one deployment window. One parameter does behave like a shared constant: a temperature offset averaged over six animals is as good as fitting each animal its own, which is what happens when the per-animal estimates are mostly estimation noise. The process-noise scale does not pool, and the obstacle there is my sampler rather than the animals.

The estimator that generalises best here is still the proprietary one, and I would rather treat that as the benchmark to beat than as a verdict on light-based geolocation. The part of this work I would defend is the protocol that let me find that out, and the part I would keep is the habit of checking whether a parameter is estimable before estimating it.

Reproducibility

Everything runs top to bottom in a single Colab notebook with no manual steps. The tag archives are fetched at run time from the ATN Data Assembly Center by stable UUID, so a cold start gets byte-identical inputs; the raw data are not redistributed. All seeds are fixed in code. Particle counts are stated with every result: Table III uses $120{,}000$ particles over ten seeds, and Tables IV, VI, VII and IX use $20{,}000$ particles over three seeds, and the readout comparison of Section IX uses five seeds. Table X uses $12{,}000$ forward particles over five seeds with 400 smoothed trajectories, and the two-filter cross-check adds $3{,}000$ backward particles on one seed. The smoother carries two self-checks that run with it: the stored-cloud forward filter is asserted bit-identical to the published one, and the backward importance ratio is asserted to be one. Table XI uses $12{,}000$ particles over five seeds, and the smoothed comparison of Section XI adds 400 trajectories on the same five seeds; the modified temperature likelihood carries its own assertion, that it is numerically identical to the published one wherever the satellite analysis has a value, so that any change in the tracks is attributable to the missing cells alone. Every parameter of the temperature model is fitted on the tuning window only. Tables XII, XIII and XIV use $12{,}000$ particles, three seeds for the sweeps and five for the controlled comparisons; the depth histogram is fitted on tuning-window Argos fixes only, the process-noise rescaling is applied through a context manager that asserts the global is restored afterwards, and the standalone likelihood used for the grid of Section XII is asserted line for line against the source of the filter it is meant to reproduce. Absolute values are therefore not comparable *between* those two groups, although every comparison *within* a table is like-for-like. Table II reports each configuration at the particle count at which it was built. Figures are regenerated by the notebook. Code, the technical report and build instructions are at https://github.com/vivianesolomon/pat-tag-geolocation. Environment: Python 3.13, NumPy 2.1.3, SciPy 1.16.3, pandas 2.2.3, Matplotlib 3.10.0, netCDF4 1.7.4.

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